

LESSON 320 (1.5 AATD)

Lesson Objective

During this lesson, the student will improve proficiency in flying VOR, localizer front course, and ILS approaches. The student will also be introduced to localizer back course approaches, GPS/WAAS approaches, and approaches utilizing DME arcs.

Briefing

- Review homework assigned in the previous lesson.
- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Student questions.

Lesson Content (Simulator)

To save time, consider skipping the instrument cockpit check and starting from the air. Use the sim's "Aircraft Attitude" menu to set their altitude, heading, and airspeed. Set the wind to be a light to moderate crosswind.

Instrument Approach Procedures

- **VOR approach with DME arc**
 - Tell the student to expect the VOR 12R @ KVRB. Have them proceed direct MLBSO. Give them a few minutes of space to run the WIRE checklist.
 - Ensure the student completes all the steps of the **WIRE** checklist:
 - **Weather**
 - **Instruments:**
 - Set both altimeters.
 - Set the correct nav source on the PFD (VLOC/GPS)
 - Load the procedure into the GPS. Load each procedure from an IAF (if ATC does not specify, choose one that makes the most sense) unless ATC states the approach will begin from radar vectors.
 - **Radios:**
 - Set COMM radios. Ensure the student sets frequencies for approach, tower, ground, and ATIS.
 - Tune/ID Nav radios. Ensure the student sets not only frequency for the navaid being used, but also any other frequencies the approach includes (for example, the MLB VOR as a part of the missed approach instructions on the ILS 09R).
 - Set the final approach course if required.
 - **Enroute brief**
 - Point out that since this approach has a FAF (Maltese cross), there is no need to calculate when to perform a final descent.
 - Emphasize that since an approach clearance has not yet been given, a descent is not

authorized. Shortly after, clear them for the approach.

- Prior to joining the final approach course, ensure the student sets airspeed to 90 knots, sets flaps to 10°, sets the MFD to the engine page, and sets the inbound course.
- Remind the student that since there is no glideslope, a proper descent rate can be maintained by setting approximately 1,800RPM (1,500RPM in the simulator) and pitching for 90 knots.
- Ensure the student makes altitude callouts at 1,000ft, 500ft, and 100ft above minimums.
- Ensure the student levels off at and holds the MDA until the MAP. Tell the student to execute the published missed approach procedure. Ensure they complete the steps for a missed approach correctly:
 - Set full power and pitch up for V_y
 - Upon reaching a positive rate of climb, retract the flaps
 - Turn to the missed approach heading/radial
 - Notify ATC

● Localizer front course

- Assign the student a heading and tell them to expect radar vectors for a LOC approach (for example, the LOC 09R at KMLB).
- Ensure the student completes all the steps of the **WIRE** checklist:
 - **Weather**
 - **Instruments:**
 - Set both altimeters.
 - Set the correct nav source on the PFD (VLOC/GPS)
 - Load the procedure into the GPS. Load each procedure from an IAF (if ATC does not specify, choose one that makes the most sense) unless ATC states the approach will begin from radar vectors.
 - Set the final approach course.
 - **Radios:**
 - Set COMM radios. Ensure the student sets frequencies for approach, tower, ground, and ATIS.
 - Tune/ID Nav radios. Ensure the student sets not only frequency for the navaid being used, but also any other frequencies the approach includes (for example, the MLB VOR as a part of the missed approach instructions on the ILS 09R).
 - **Enroute brief**
- Provide alternate missed approach instructions (for example, heading 220 and 2,000ft). Explain that these instructions apply to all subsequent approaches unless otherwise stated.
- Fail the glideslope using the Instruments menu.
- Give the student a vector to final along with an approach clearance. This is a good time to ensure airspeed is 90 knots, extend the flaps to 10°, and set the MFD to the engine page.

Instrument Rating Course

- Ensure the student makes altitude callouts at 1,000ft, 500ft, and 100ft above minimums.
- Explain that since there is no glideslope, a proper descent rate can be maintained by setting approximately 1,800RPM (1,500RPM in the simulator) and pitching for 90 knots.
- Ensure the student levels off at and holds the MDA until the MAP. Tell the student to execute the published missed approach procedure. Ensure they complete the steps for a missed approach correctly:
 - Set full power and pitch up for V_y
 - Upon reaching a positive rate of climb, retract the flaps
 - Turn to the missed approach heading/radial
 - Notify ATC

● ILS

- Assign the student a heading and tell them to expect radar vectors for an ILS approach (for example, the ILS 09R at KMLB).
- Ensure the student completes all the steps of the **WIRE** checklist:
 - **Weather**
 - **Instruments:**
 - Set both altimeters.
 - Set the correct nav source on the PFD (VLOC/GPS)
 - Load the procedure into the GPS. Load each procedure from an IAF (if ATC does not specify, choose one that makes the most sense) unless ATC states the approach will begin from radar vectors.
 - **Radios:**
 - Set COMM radios. Ensure the student sets frequencies for approach, tower, ground, and ATIS.
 - Tune/ID Nav radios. Ensure the student sets not only frequency for the navaid being used, but also any other frequencies the approach includes (for example, the MLB VOR as a part of the missed approach instructions on the ILS 09R).
 - Set the final approach course.
 - **Enroute brief**
- Give the student a vector to final along with an approach clearance. This is a good time to ensure airspeed is 90 knots, extend the flaps to 10° , and set the MFD to the engine page.
- Explain that setting around 1,800RPM in the aircraft (1,500RPM in the sim) will provide a roughly -500fpm descent rate at 90 knots.
- Ensure the student makes altitude callouts at 1,000ft, 500ft, and 100ft above minimums.
- Ensure the student executes their missed approach upon reaching the DA:
 - Set full power and pitch up for V_y
 - Upon reaching a positive rate of climb, retract the flaps
 - Turn to the missed approach heading/radial
 - Notify ATC

- **Localizer back course**
 - Assign the student a heading and tell them to expect radar vectors for a LOC approach (for example, the LOC 09R at KMLB).
 - Ensure the student completes all the steps of the **WIRE** checklist:
 - **Weather**
 - **Instruments:**
 - Set both altimeters.
 - Set the correct nav source on the PFD (VLOC/GPS)
 - Load the procedure into the GPS. Load each procedure from an IAF (if ATC does not specify, choose one that makes the most sense) unless ATC states the approach will begin from radar vectors.
 - **Radios:**
 - Set COMM radios. Ensure the student sets frequencies for approach, tower, ground, and ATIS.
 - Tune/ID Nav radios. Ensure the student sets not only frequency for the navaid being used, but also any other frequencies the approach includes (for example, the MLB VOR as a part of the missed approach instructions on the ILS 09R).
 - Set the final approach course. Remind them that on back course approaches they must set the front course.
 - **Enroute brief**
 - Give the student a vector to final along with an approach clearance. This is a good time to ensure airspeed is 90 knots, extend the flaps to 10°, and set the MFD to the engine page.
 - Ensure the student makes altitude callouts at 1,000ft, 500ft, and 100ft above minimums.
 - Explain that since there is no glideslope, a proper descent rate can be maintained by setting approximately 1,800RPM (1,500RPM in the simulator) and pitching for 90 knots.
 - Ensure the student levels off at and holds the MDA until the MAP. Ensure they complete the steps for a missed approach correctly:
 - Set full power and pitch up for Vy
 - Upon reaching a positive rate of climb, retract the flaps
 - Turn to the missed approach heading/radial
 - Notify ATC
- **GPS/WAAS RNAV approach**
 - Tell the student to proceed direct to an IAF for an RNAV approach (for example, the RNAV 27L @ KMLB). Proceed to this fix via the PROC page.
 - Ensure the student completes all the steps of the **WIRE** checklist:
 - **Weather**
 - **Instruments:**
 - Set both altimeters.
 - Set the correct nav source on the PFD (VLOC/GPS).

Instrument Rating Course

- Load the procedure into the GPS. Load each procedure from an IAF (if ATC does not specify, choose one that makes the most sense) unless ATC states the approach will begin from radar vectors.
- Radios:
 - Set COMM radios. Ensure the student sets frequencies for approach, tower, ground, and ATIS.
 - Tune/ID Nav radios. Ensure the student sets not only frequency for the navaid being used, but also any other frequencies the approach includes (for example, the MLB VOR as a part of the missed approach instructions on the ILS 09R).
 - Set the final approach course if required.
- Enroute brief
 - Set the weather to allow for a landing.
 - Prior to reaching the IAF, give the student an approach clearance.
 - Point out the G430 prompts to turn just prior to reaching fixes on the approach so as not to overshoot.
 - If autopilot is to be used, engage GPSS mode by pressing NAV twice to ensure this happens automatically.
 - Prior to final approach course intercept and when cleared for the approach, ensure airspeed is 90 knots, extend the flaps to 10°, and set the MFD to the engine page.
 - Explain that setting around 1,800RPM in the aircraft (1,500RPM in the sim) will provide a roughly -500fpm descent rate at 90 knots.
 - Ensure the student makes altitude callouts at 1,000ft, 500ft, and 100ft above minimums.
 - Remind the student of the 91.175(c) criteria required to land. Once these criteria are met, extend the flaps and land.

Post-Brief

- Student Self Critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 321.
- Assign Homework:
 - Review procedures for ILS and GPS approaches.
 - Review current NOTAMs for the expected approaches.

Completion Standards

At the conclusion of this lesson, the student should:

1. Have a basic understanding of the following approaches: DME arc, localizer back course

- approach, and GPS/WAAS approach.
2. Be able to maintain specific descent rates, altitudes, and navigation tracking during each approach.
 3. Demonstrate glide slope tracking, using pitch and power changes to control descent rates and airspeed during the ILS and/or WAAS instrument approach procedures.